

Question:
What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

- Answer Choices:
- A. 2.48×10^{-8} cm
 - B. 5.92×10^7 cm/sec
 - C. 8.66×10^{-6} cm
 - D. 3.14×10^{-5} cm
 - E. 9.81×10^{-4} cm
 - F. 5.27×10^{-9} m
 - G. 1.23×10^{-7} cm
 - H. 4.14×10^{-10} m
 - I. 6.625×10^{-27} erg sec
 - J. 0.512×10^6 eV

Reasoning + Answer:

- <think>
- First , convert the kinetic energy from eV to joules . $1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$.
 - The energy of a particle is given by $E = \frac{h^2}{2m\lambda^2}$ from the de Broglie relation , where h is Planck's constant , m is the mass of the electron , and λ is the wavelength .
 - The mass of an electron $m = 9.109 \times 10^{-31} \text{ kg}$, and Planck's constant $h = 6.626 \times 10^{-34} \text{ Js}$.
 - Rearrange the de Broglie relation to solve for λ , the de Broglie wavelength : $\lambda = \sqrt{2mE/h^2}$.
 - Substitute $E = 1.602 \times 10^{-19} \text{ J}$, $m = 9.109 \times 10^{-31} \text{ kg}$, and $h = 6.626 \times 10^{-34} \text{ Js}$ into the formula to find λ .
 - Calculate λ , remember to convert from m to cm at the end by multiplying by 100 .

Plugging the numbers into the equation : $\lambda = \sqrt{(2 * 9.109 \times 10^{-31} \text{ kg} * 1.602 \times 10^{-19} \text{ J}) / (6.626 \times 10^{-34} \text{ Js})^2} \approx 1.23 \times 10^{-7} \text{ cm}$.

</think>

<answer> 1.23×10^{-7} cm</answer>